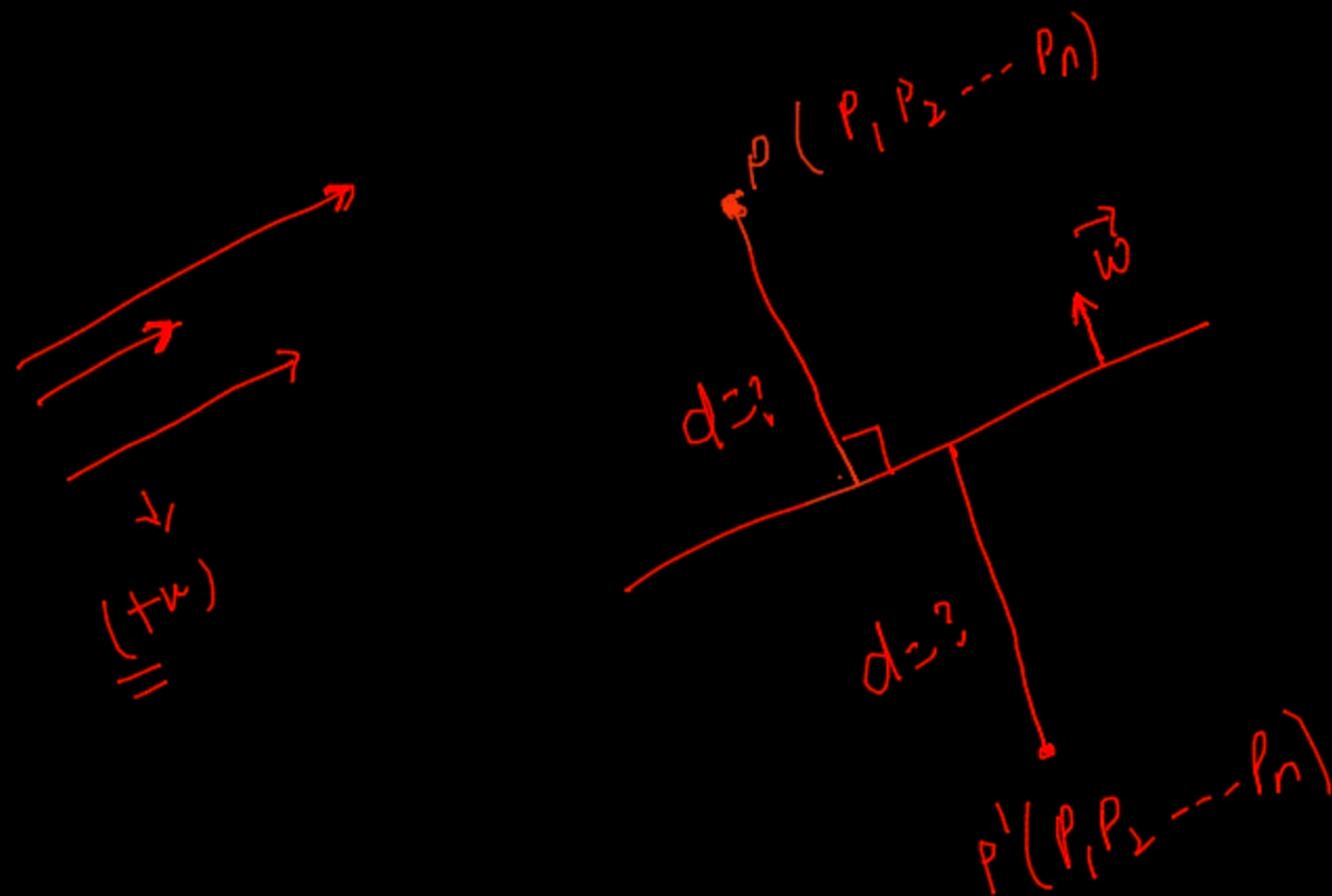




distance of a point from a plane

$$d = \frac{w^T \cdot P}{\|w\|}$$

Here P and w are on the same side

$$d = \frac{w^T \cdot P}{\|w\|} \rightarrow (+w) \rightarrow \boxed{+}$$

$$d' = \frac{w^T \cdot P'}{\|w\|} \Rightarrow (-w) +$$

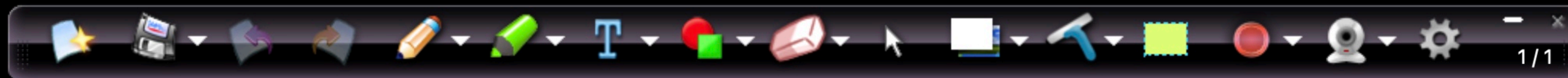
2D

$$\text{line} \rightarrow \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} [x_1, x_2] + w_0$$

w_0



Talking: Rahul Tripathi



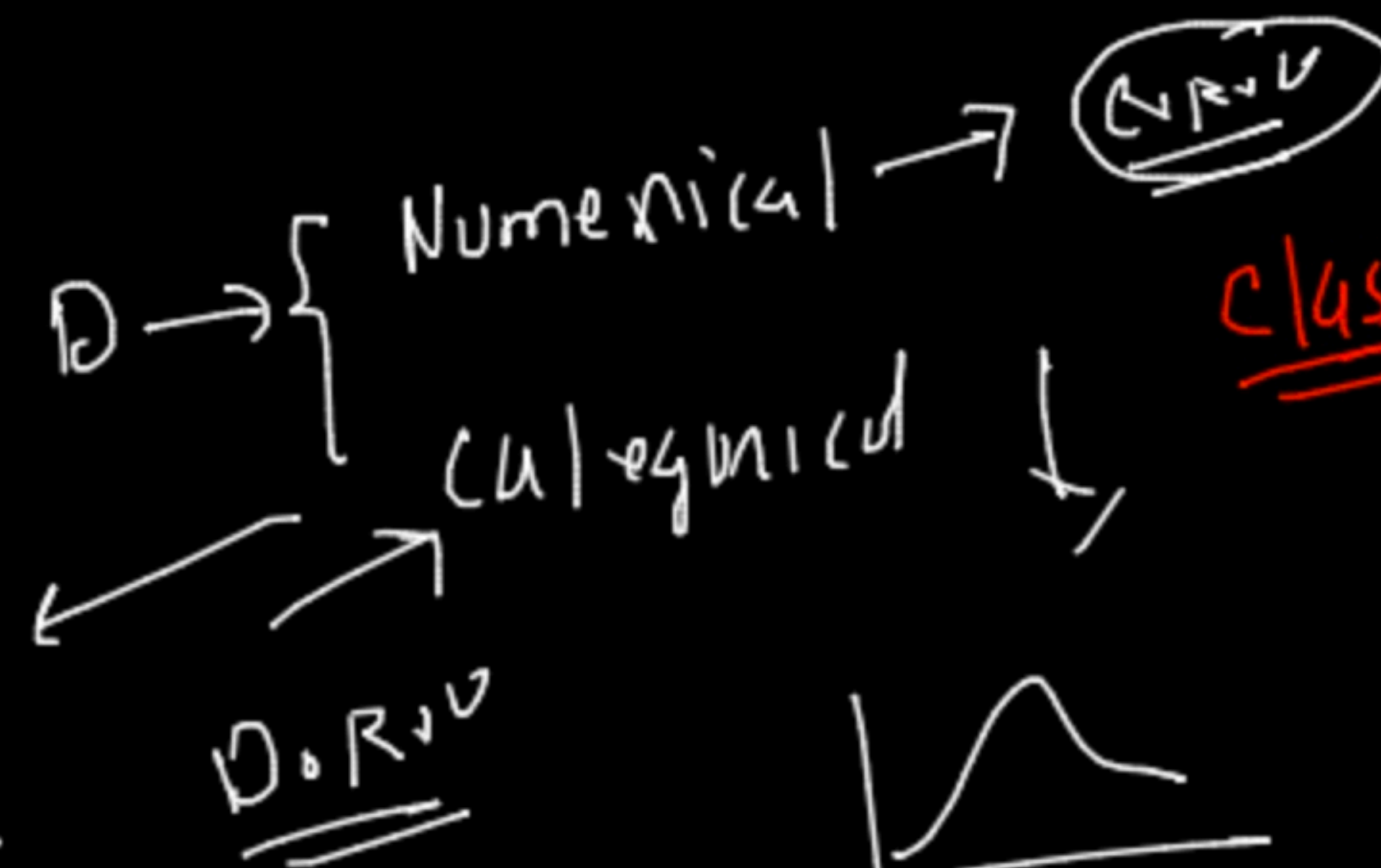
IML → ☒ → Prob & Stats

↳ Random Variable



age = 24

↑
variable



age 24

toss a coin

But $X = \{H, T\}$

↑
finite

Random Variable
 $X = H/T$

classroom
'100' →

height = {144, 145, ... ∞}

C.R.V.

$X = \{H, T\}$

~~X~~ = {1, 2, 3, 4, 5, 6}

Discrete Random variable

Continuous Random variable

Analysis

Talking: Rahul Tripathi

$$X = \{1, 2, 3, 4, 5, 6\} \leftarrow$$

$$P(X=1) = \frac{1}{6}$$

Experiment

$$\{1, \boxed{2}, 3, \boxed{4}, 5, \boxed{6}\}$$

↑

$$\frac{1}{6}$$

$$P(X \text{ is a even number}) = \frac{3}{6} \Rightarrow \frac{1}{2}$$

$$P(X=2) = \frac{1}{6} \Rightarrow$$

↑ ↑

$$\frac{\text{\# time event}}{\text{total \# of event}}$$

(X=2)

Talking: Rahul Tripathi

outliers
↑

$y = \text{height}$

$\{122, 146.4, 134.5, \dots, 12.25\}$
↑

'outliers'
==

mean →

Talking: Rahul Tripathi

Population and Sample

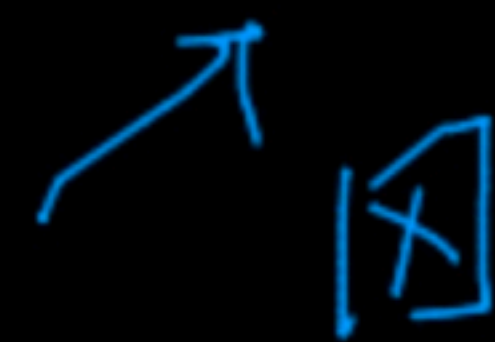


of samples



Sample mean $\bar{X}$
($\bar{X} \approx \mu$)

$$\text{Avg } h_i = \frac{h_1 + h_2 + \dots + h_{TB}}{TB} \Rightarrow$$



Sample

proof

$$S_3 = \{ \dots \}$$



$\bar{X}_3$

$$S_2 = \{ \dots \} \rightarrow \bar{X}_2$$



$$S_1 = \{ \dots \}$$



$$\frac{\sum_{i=1}^n h_i}{\text{len}(S_1)} \Rightarrow \bar{X}_1$$

Talking: Rahul Tripathi

Histogram
↑

Salary | Age | h

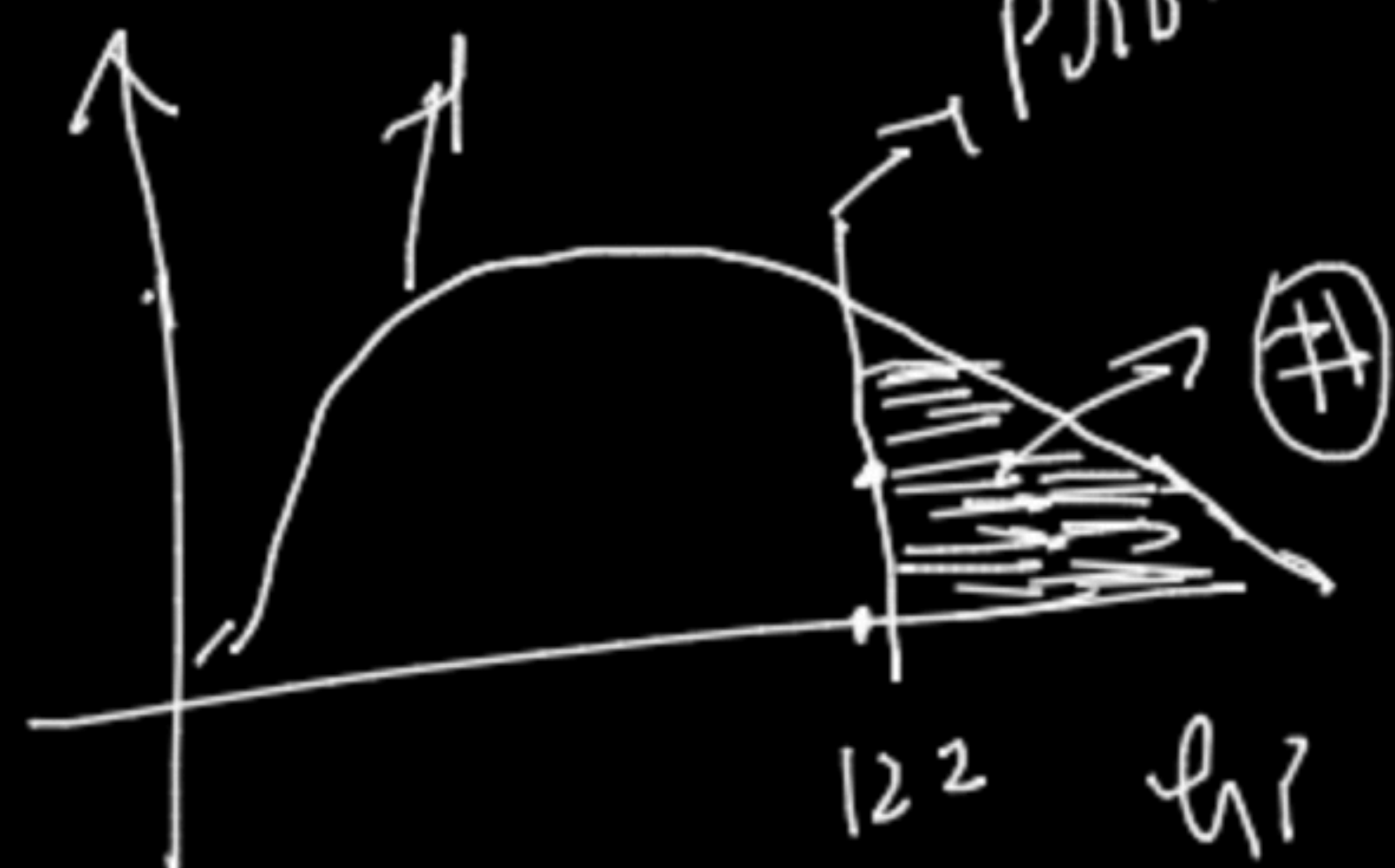
Area under A

$$\int_{-\infty}^{\infty} f(x) dx$$

↑

$f(x)$

Prob. density fn



$$P(X > 122) = ?$$

↪

tossing a coin

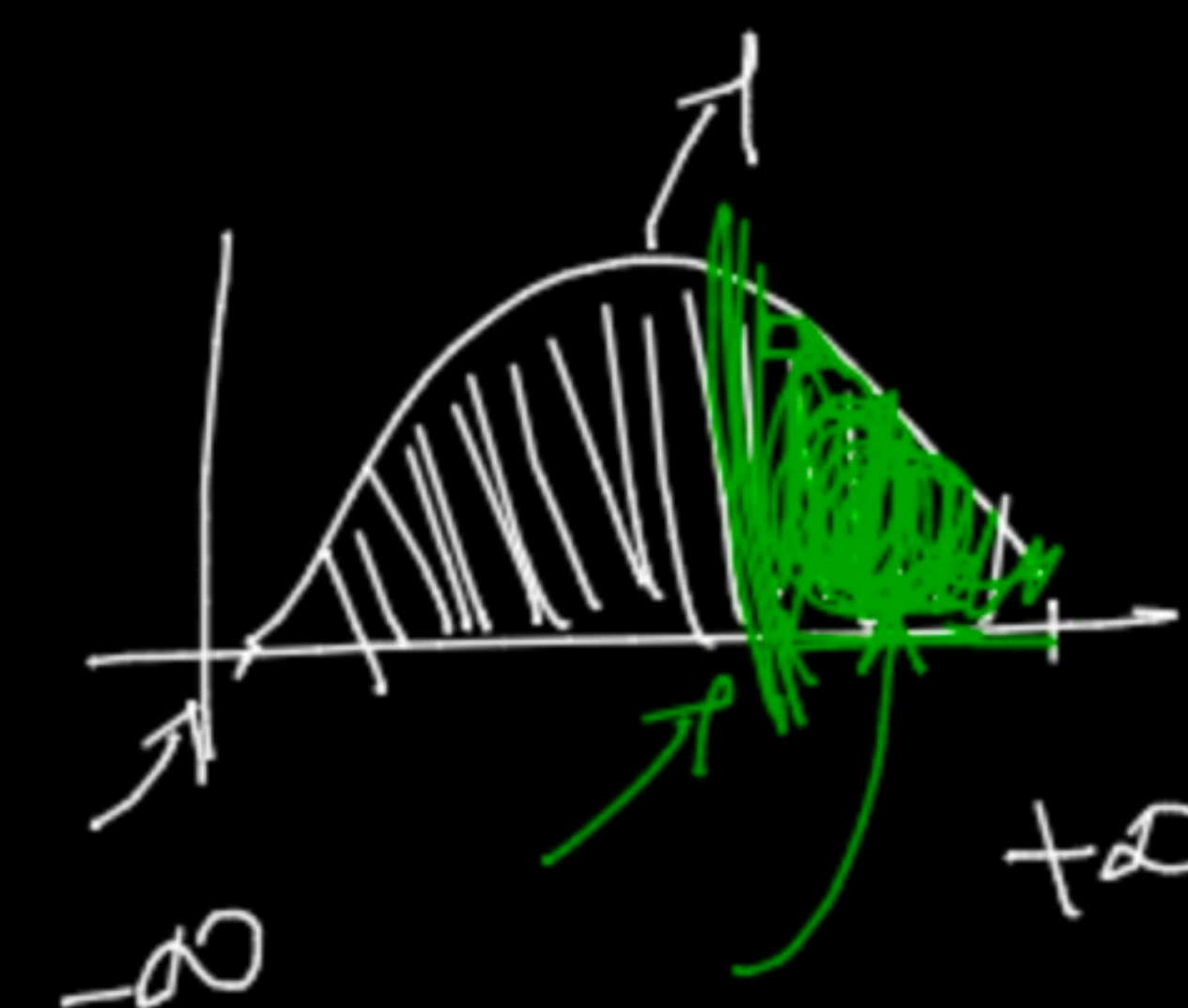
D.R.V

X	H	T
P(X)	1/2	1/2

P.D.T.

$$\int_{-\infty}^{\infty} f(x)$$

11



← X

(initial)

C.R.V

P(X)

↪

$$P(X = 122)$$

↪ (X)

$$\int_{-\infty}^{\infty} f(x) dx$$

Talking: Rahul Tripathi

→ Gaussian distribution

→ Normal distribution

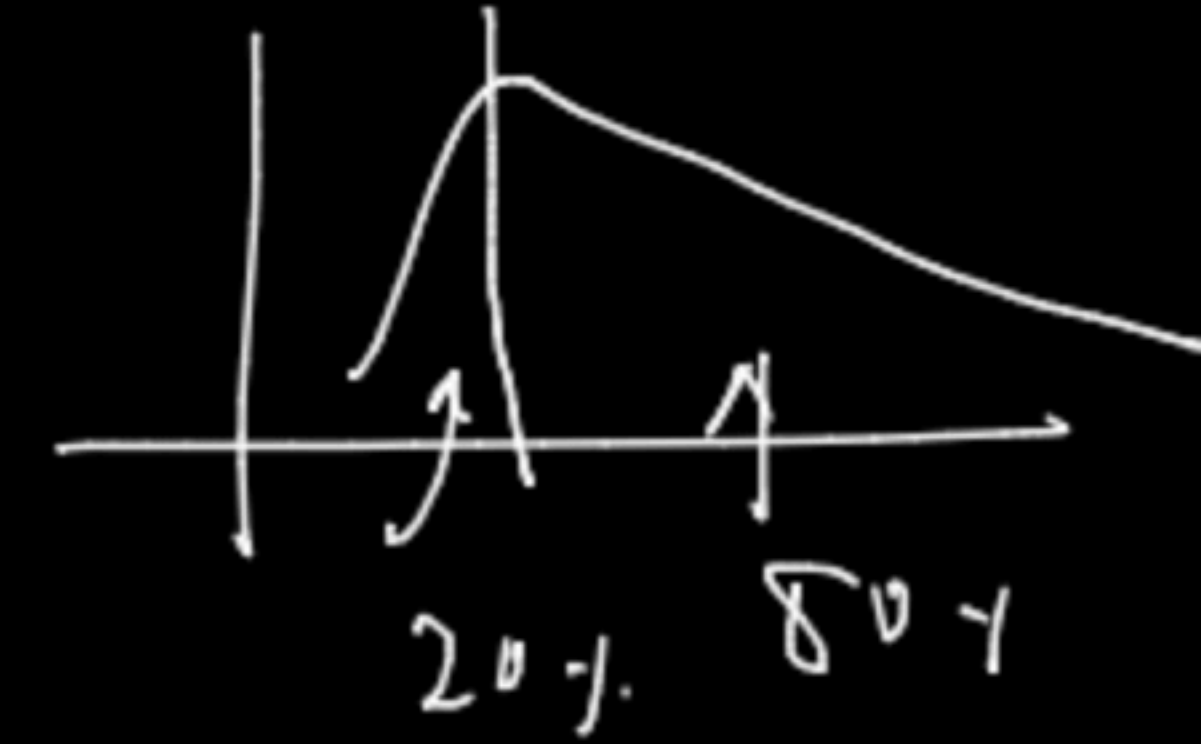
$$X \sim N(\mu, \sigma^2)$$

shape of curve variance

$$X \Rightarrow \{ \text{-----} \}$$

$$\uparrow$$
$$\text{mean}(X) = ?$$

$$\text{var}(X) = ?$$



Talking: Rahul Tripathi

PDF

$$P(X=x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{ \frac{-(x-\mu)^2}{2\sigma^2} \right\}$$

$$X = 145$$

$$\text{mean} = \mu = 144$$

$$\sigma^2 = ?$$

$$P(X=145) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{ \frac{-(145-144)^2}{2\sigma^2} \right\} \rightarrow \textcircled{?}$$

$$\mu = 0$$

$$\sigma^2 = 1$$

$$\sigma = 1$$

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$$

$$= \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{(x)^2}{2}\right\}$$

$$= \left(\frac{1}{\sqrt{2\pi}}\right) \exp\left\{-\frac{1}{2}x^2\right\}$$

$$\Rightarrow \frac{1}{\sqrt{4}} \exp\{-x^2\} \rightarrow \text{plot}$$

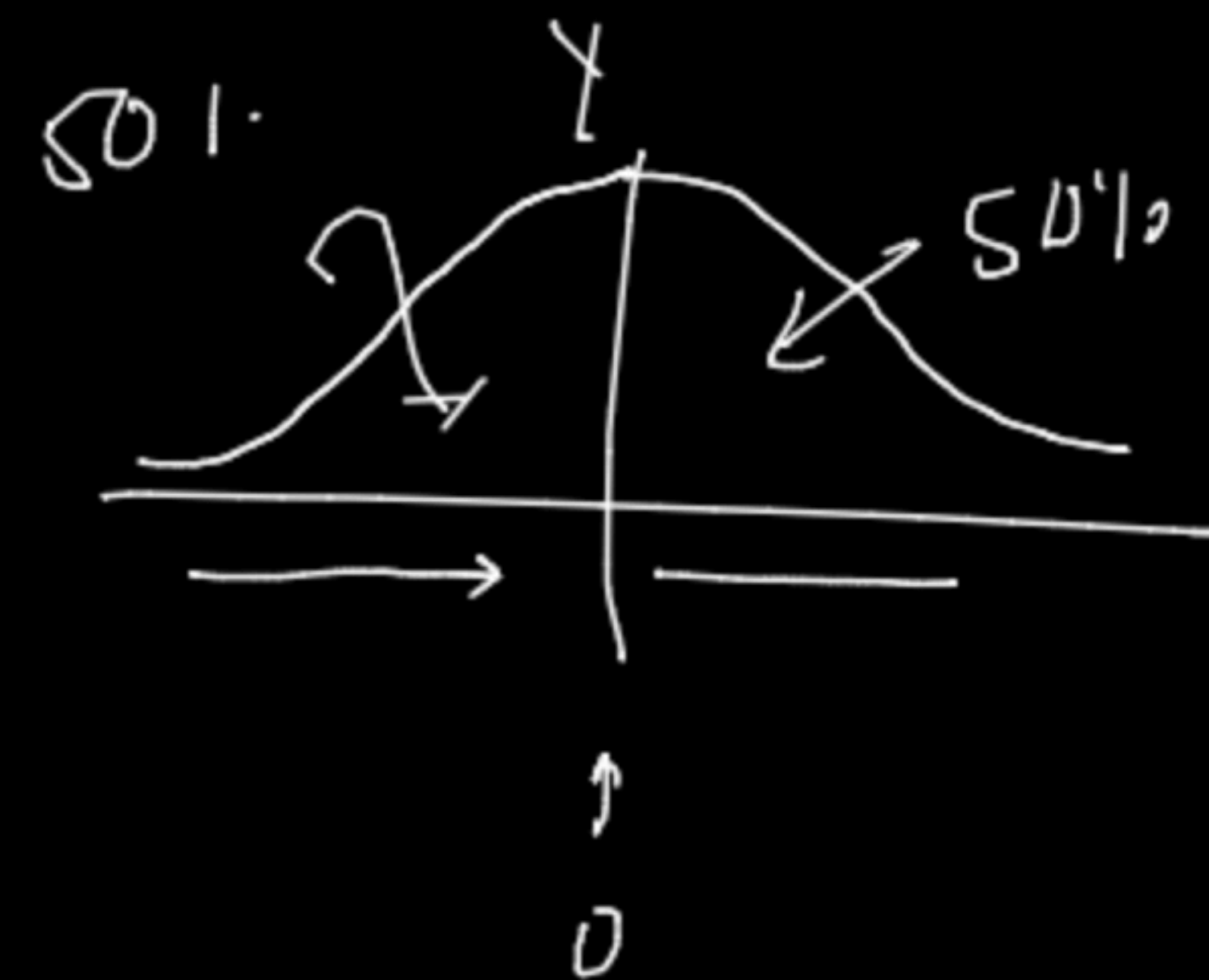
conclusion

① $\rightarrow$ x moves away from mean

$\rightarrow$ y reduce

② symmetric

③ bell shape curve



④ $x \rightarrow$ mean
 $\rightarrow y \uparrow$

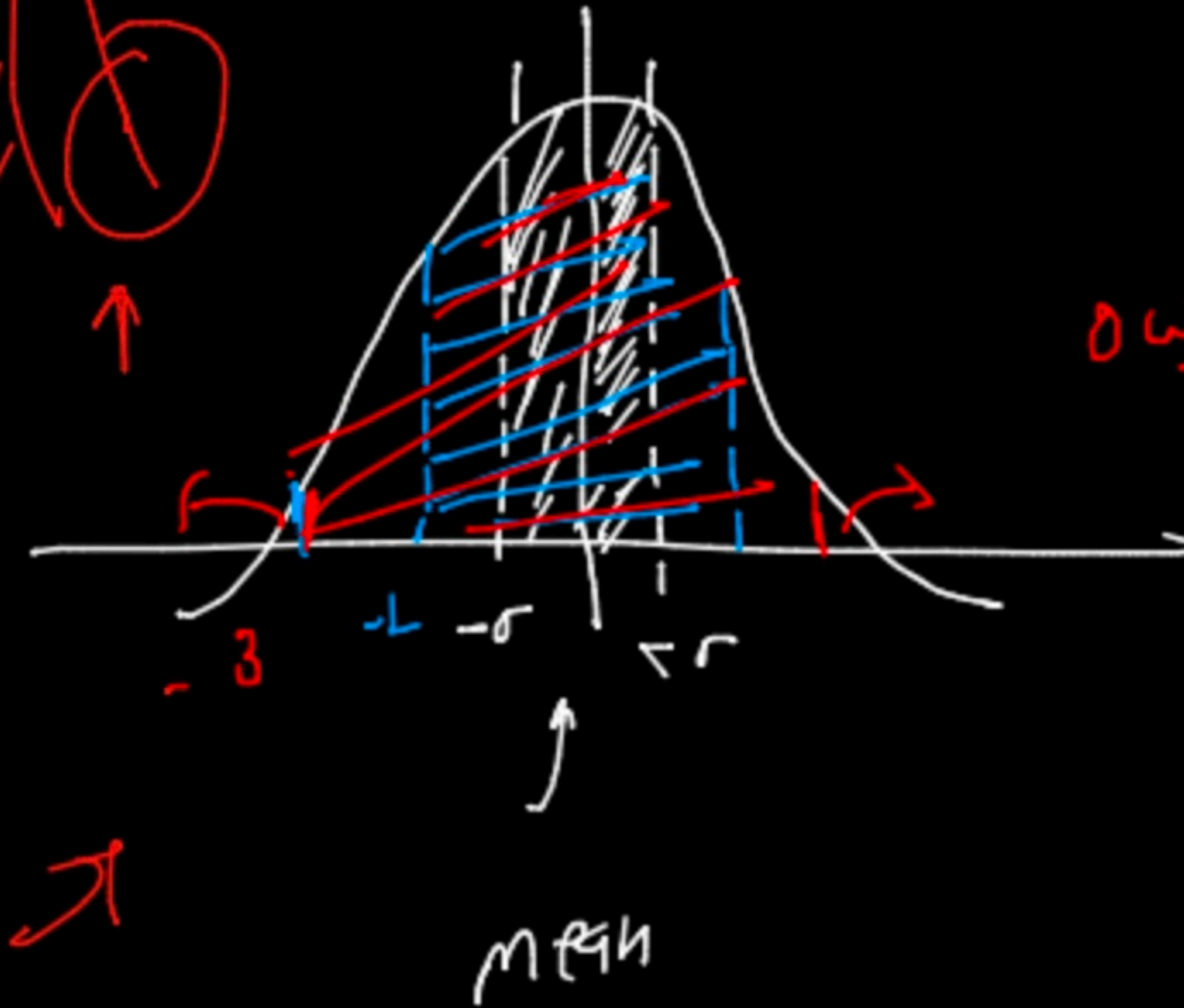
Talking: Rahul Tripathi

→ 68-95-99 Rule

μ → mean

Normal distribution →

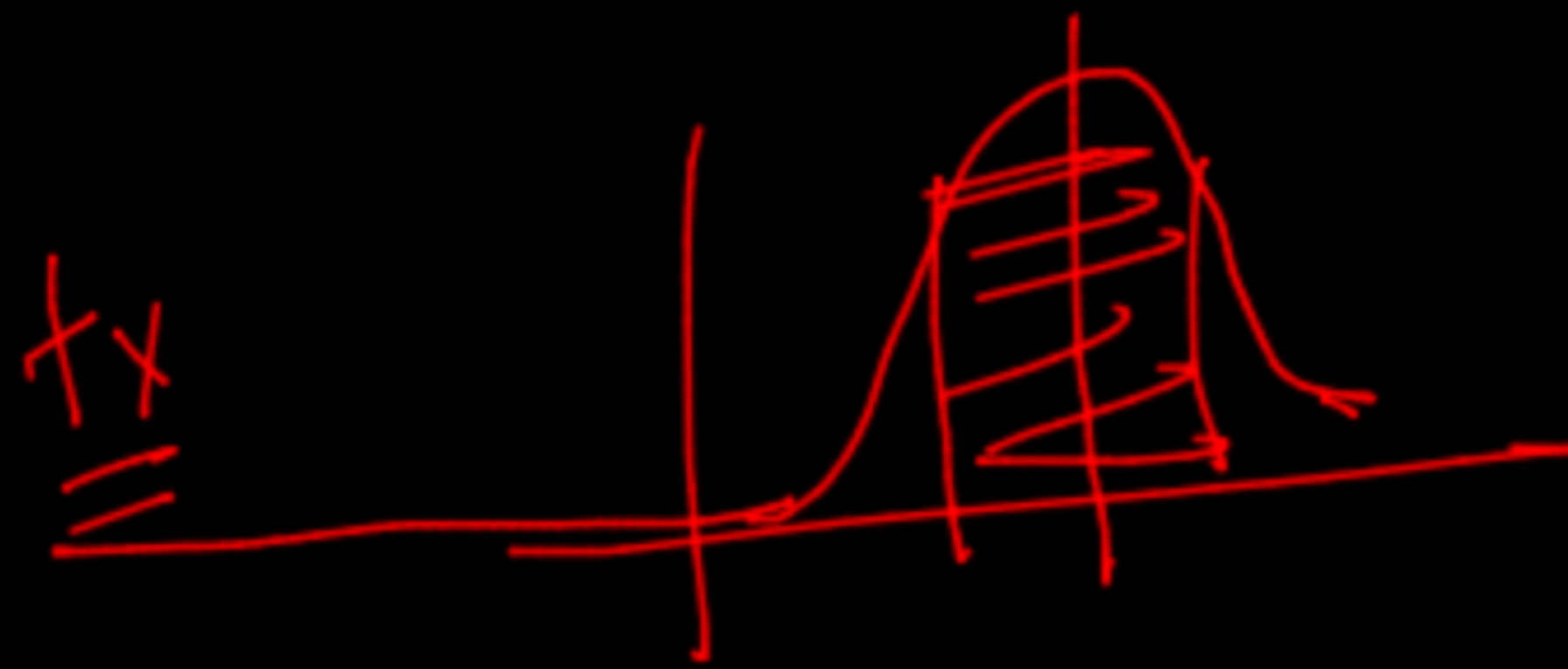
84%



eg.

TCS info

$\begin{cases} S \rightarrow ? \\ M \rightarrow ? \\ XL \end{cases}$



b/w $-\sigma$ to $\sigma \rightarrow [68\%]$ ✓

-2σ to $+2\sigma \rightarrow 95\%$ ✓

-3σ to $+3\sigma \rightarrow 99.7\%$ ✓

begin

10

81%

code

$\begin{pmatrix} h_1 & \dots \end{pmatrix}$
 $\begin{pmatrix} h_1 & \dots \end{pmatrix}$
 $\begin{pmatrix} h_1 & \dots \end{pmatrix}$

$h \rightarrow NO$ → 68%



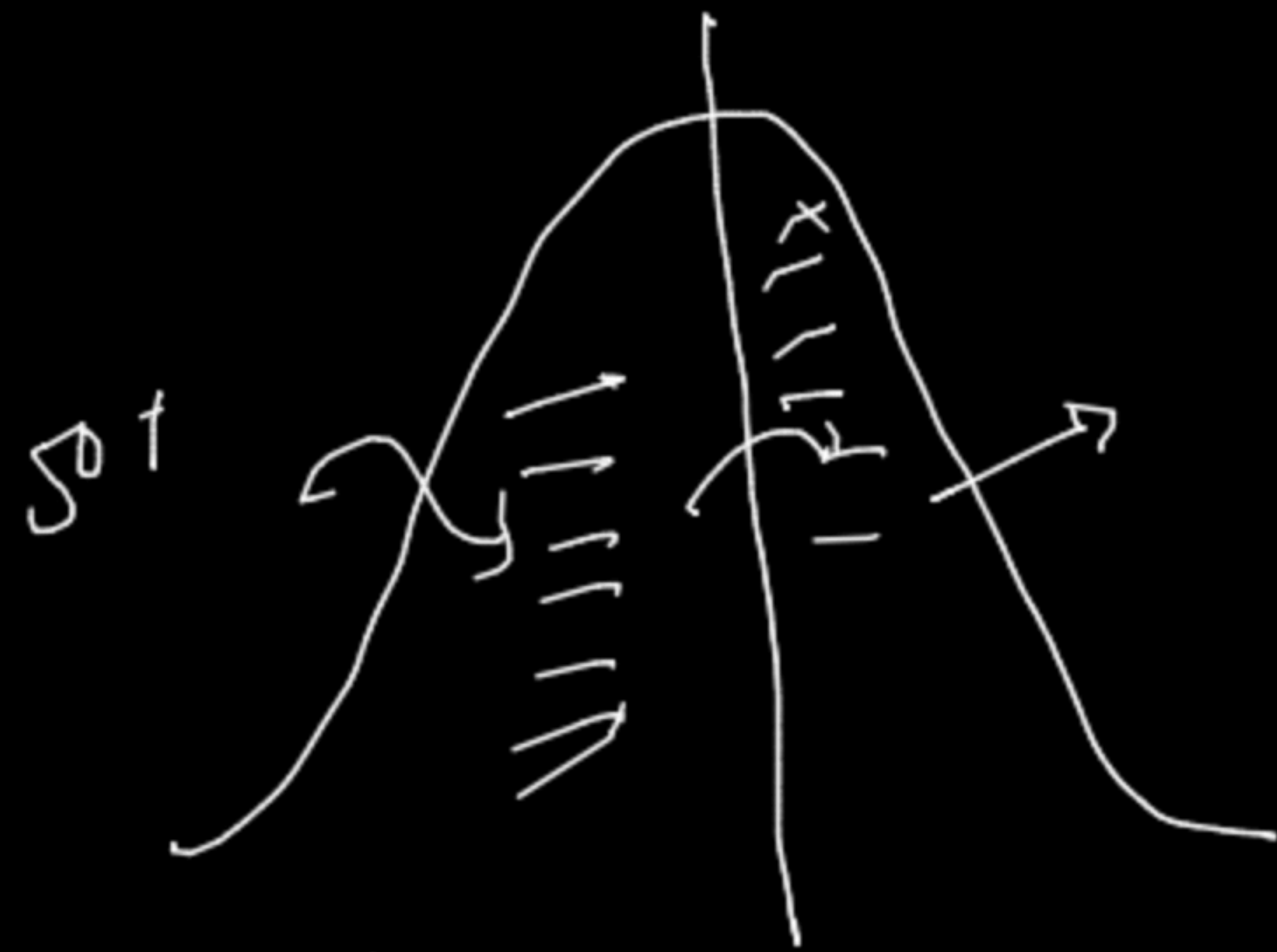
Talking: Rahul Tripathi

Symmetrical distribution

Skewness

Kurtosis

tells the shape of PDF



ML

Exam

TI $\rightarrow$ 68.12
T2 $\rightarrow$ -
T3 $\rightarrow$ 3

Exam

Lean

(Skewness)

ML

(Data)
ML

ND

ML

(Data)

X

ET

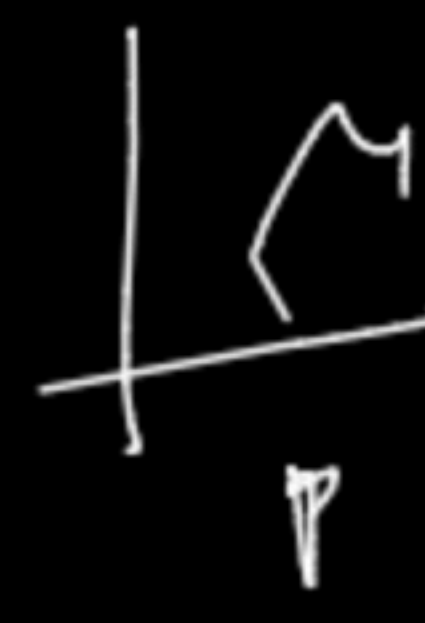


ND

(Rise)

log()

ML



Talking: Rahul Tripathi

$\Rightarrow$ Standard Normal distribution

$$Z \sim N(0, 1)$$

$$h = [x_1 \ x_2 \ x_3 \ \dots \ x_n]$$

$\uparrow$
mean(h)

std = ?

transforming

$$\frac{x_i - \mu}{\sigma}$$

$$x' = \{0.1, 0.2, 0.1, \dots\}$$

$$\{45, 144, \dots\}$$

$$\text{mean} \Rightarrow 148$$

$$\text{std} \Rightarrow 0.5$$

$$\{0.7\}$$



plot $\uparrow$ ND

Standardize

Talking: Rahul Tripathi

CLT

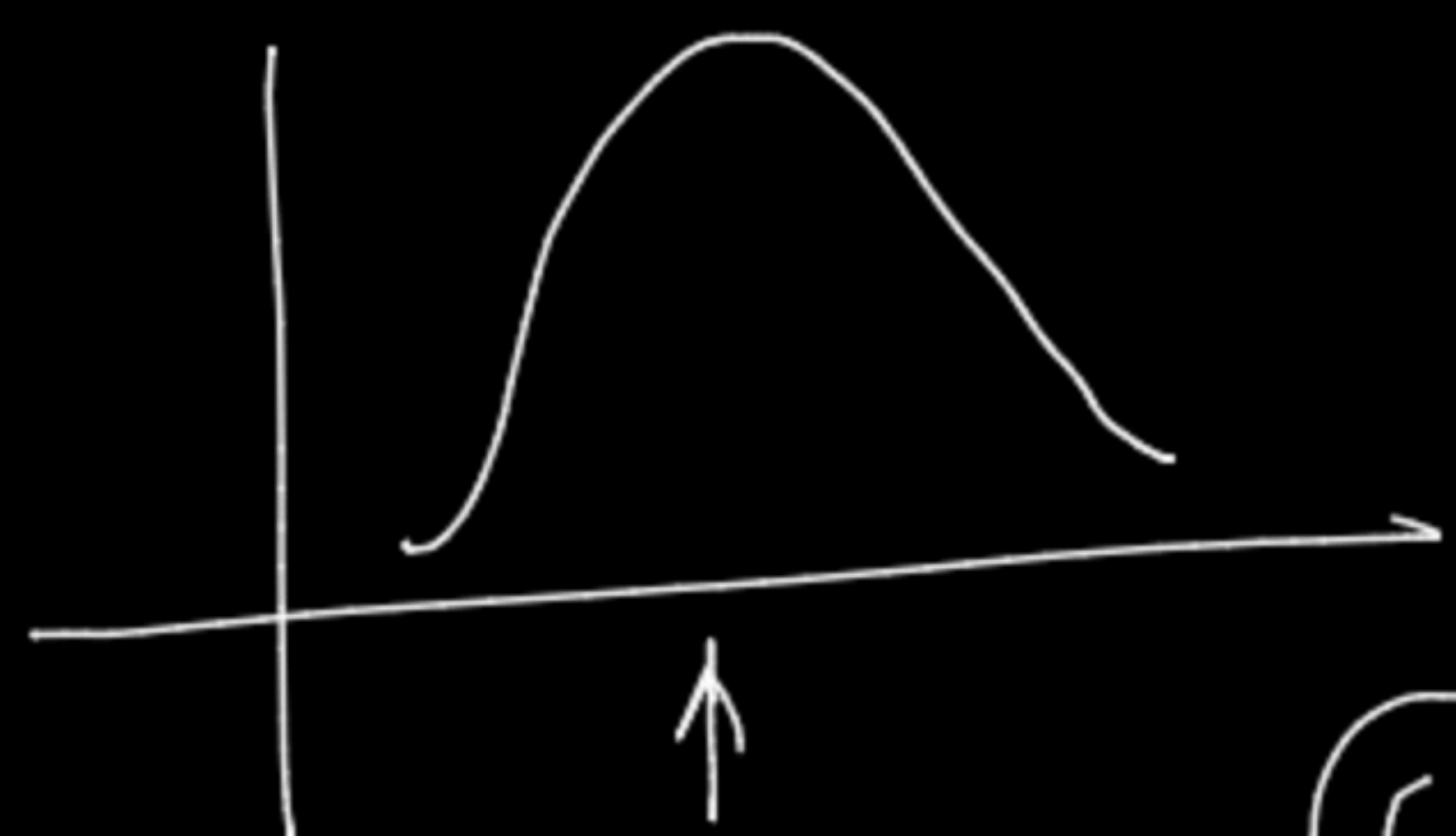
Sampling distribution



μ

σ^2

$N.D$
 $F(x)$



Central limit theorem

CLT

Random sample $\rightarrow$ of size n (Let $n=30$)

$S_1 \rightarrow \bar{x}_1$

$S_2 \rightarrow \bar{x}_2$

$S_3 \rightarrow \bar{x}_3$

$\vdots$
 $S_k \rightarrow \bar{x}_k$

$\bar{x}_1 = N\left(\mu, \sqrt{\frac{\sigma^2}{n}}\right)$ as $n \rightarrow \infty$

μ σ^2 n

plot all the means

$\bar{x} = \{\bar{x}_1, \bar{x}_2, \dots\}$

Talking: Rahul Tripathi